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Direct analysis of dissolved CO_2 in coastal waters: development and validation of a simple method

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ABSTRACT

The partial pressure of CO_2 ($p(CO_2)$) dissolved in seawater is often indirectly determined through a mathematical calculation involving two variables of the carbonate system, which requires the use of expensive equipment, materials, and reagents, and is usually demanding in terms of human resources. In this study, we present a simple, low-cost method for directly measuring dissolved CO_2 ($p(CO_2)$) in coastal waters using a small gas equilibrator coupled with a non-dispersive infrared (NDIR) detector in a closed-loop system. The cost of this methodology can be further reduced by utilizing a low-cost NDIR detector and inexpensive materials. Its analytical performance was validated against the conventional indirect approach based on pH and dissolved inorganic carbon (DIC), showing high accuracy (91 %), excellent linearity ($r^2 = 0.998$), and a quantification limit (~21 μatm) approximately 20 times lower than the average surface seawater $p(CO_2)$. The method proved robust to typical variations in salinity (15–45 PSU) and temperature (20–25 °C), though a new calibration is recommended for salinities below 15 PSU. Application to samples from a tropical coastal lagoon and adjacent oceanic area showed statistically comparable results with both methodologies. This approach could enable direct measurement in the field, thereby reducing labor, equipment, and material costs. This low-cost approach enhances in situ $p(CO_2)$ monitoring, offering significant potential for long-term ocean acidification studies in resource-limited settings.

1. Introduction

The atmospheric concentration of carbon dioxide (CO_2) has been increasing continuously since the beginning of the twentieth century due to rapid population growth and human activities, of which the emissions increase due to the burning of fossil fuels stand out (Mardani et al., 2019). The greenhouse effect promoted by the atmospheric enrichment of CO_2 has increased the global temperature by ~1 °C since pre-industrial times, causing innumerable consequences, such as increasing hydrometeorological events (droughts and floods) (IPCC, 2021) and the reduction of the capacity of CO_2 sinks (such as forests and oceans) to absorb CO_2 due to their saturation (WMO, 2022).

Among these CO_2 sinks, coastal ecosystems such as estuaries, salt marshes, mangroves, and seagrass meadows play a very important role in the global carbon cycle. Despite covering a small fraction of the ocean surface, they contribute significantly to carbon sequestration, acting as dynamic interfaces between terrestrial and marine carbon fluxes. However, these environments are also highly vulnerable to anthropogenic pressures and climate change, making the monitoring of carbon dynamics in coastal areas a global priority (Duarte et al., 2005; Alongi, 2014).

The absorption of atmospheric CO_2 in the oceans causes acidification because it affects the balance in the carbonate system in aqueous media (Equations (1) and (2)), decreasing the availability of CO_3^{2-} and

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increasing the amount of hydrogen ions (Doney et al., 2009). The reduction of the availability of carbonates affects the calcification of many organisms and can have adverse effects on others (Gatusso et al., 1998).

$$CO_{2(g)} + H_2O \rightleftarrows H^+ + HCO_3^- \quad (1)$$

$$HCO_3^- \rightleftarrows H^+ + CO_3^{2-} \quad (2)$$

Traditionally, the partial pressure of CO_2 in seawater ($p(CO_2)$) is not measured directly, but rather estimated mathematically from two carbonate system parameters (such as total alkalinity (TA), dissolved inorganic carbon (DIC), or pH) using thermodynamic models. Although widely used, these indirect methods present several limitations: (i) they require sophisticated and expensive instruments (e.g., UV–Vis spectrophotometers), and (ii) they are labor-intensive, often requiring highly trained personnel and laboratory conditions (Izumi et al., 2022; Job et al., 2023). As a result, continuous or high-frequency $p(CO_2)$ monitoring in coastal areas (especially in developing countries) remains limited. These challenges underscore the need for simplified and accessible methodologies that can directly measure $p(CO_2)$ in discrete samples.

To overcome these barriers, we present a novel and low-cost methodology for directly measuring $p(CO_2)$ in coastal waters, addressing the practical limitations of existing approaches. Its main novelty is the coupling of a small gas equilibrator with a non-dispersive infrared (NDIR) detector (LI-850), in a closed-loop system. Validated against conventional techniques, our method achieves high accuracy (91 %) and excellent linearity ($r^2 = 0.998$), with robustness across typical coastal conditions. This approach was tested in small (20 mL) samples from a coastal lagoon, and may be easily used in field conditions, thus reducing labour, equipment, and materials costs. The cost of this methodology can be further reduced by utilizing existing, low-cost NDIR detectors and materials. This accessible, portable solution has the potential to ease $p(CO_2)$ monitoring, enabling broader participation in critical ocean acidification and carbon cycle research.

2. Methodology

In this study, two methodologies for analyzing $p(CO_2)$ were employed: the direct measurement method developed, standardized, and validated here, and the indirect measurement method widely used as a reference method (Dickson et al., 2007).

2.1. Direct measurement of $p(CO_2)$

For the direct measurement of $p(CO_2)$, we used a closed coupling between a gas equilibrator (RAD-H_2O®, Durridge) and a CO_2 detector (LICOR-850®) (Fig. 1). The LICOR-850 (or LI-850) consists of a suction pump and an analysis cell that measures CO_2 and water vapor concentrations in the air using a Non-Dispersive Infrared detector (NDIR). As the infrared signal of water vapor interferes with that of CO_2, the equipment features a calibration function to distinguish between them (LI-COR, 2023). A Drierite desiccator tube was added to retain the abundant water vapor from the seawater samples. Airflow, CO_2, and H_2O absorbances, cell pressure and temperature were recorded every 0.5 s through the LI-850 control software.

The gas equilibrator (Fig. 2) is a 40 mL borosilicate bottle with a septum cap. These bottles are designed to measure ^{222}Rn, a radioactive noble gas. In each bottle, two 10 mL seawater aliquots were added using a micropipette (Across® OHAUS AO-10ML), resulting in a total volume of 20 mL of seawater and 20 mL of air (headspace).

The bottle was sealed with a stainless-steel aerator cap with two connection ports (for the air inlet and outlet) (Fig. 2). Gas was bubbled through the sample in the gas equilibrator using a glass tube (frit) that reached the bottom of the bottle, thereby carrying the sample's $CO_{2(g)}$ through the outlet port of the aerator cap. The sample CO_2 was transported through the system by the LI-850 suction pump at 0.70 L min^{-1} and dried in a Drierite tube before measurement in the LI-850. After measurement, the gas was reinjected into the sample through the inlet port of the aerator cap. The hose connected to the inlet port features a one-way valve, ensuring air flow in only one direction, thereby creating a closed loop (Fig. 1).

2.1.1. Sample analysis

Water sample analysis consists of 2 steps: air evacuation and sample measurement. The whole procedure is monitored with the LI-850.

Step 1: Air evacuation

Since CO_2 concentrations in laboratory air are typically higher than 400 ppm, the procedure was conducted under a nitrogen atmosphere. An empty bottle was placed in the gas equilibrator to evacuate ambient air from the system using high-purity nitrogen (99.90 %) at a flow rate <1 L min^{-1}, continuously expelled outside (open system; Fig. 1). As a result, the CO_2 baseline stabilized and was considered the measurement blank.

Step 2: Sample measurement

Once the measurement blank was established, the nitrogen flow was stopped. The system configuration was quickly changed to measure the sample: the hose was disconnected from the nitrogen tank and connected to the outlet port of the LI-850 to close the system (Fig. 1). Then, the empty bottle in the gas equilibrator was substituted with the sample bottle, and the LI-850 pump was turned on. As this is now a closed-loop system, the CO_2 in the sample bottle is continuously mixed with the nitrogen inside the system until it reaches homogenization. The LI-850 recorded CO_2 measurements every 0.5 s for 20 min.

2.1.2. Data analysis

An example of the measurement record is shown in Fig. 3. During air evacuation, the flow is ~1 L min^{-1} (the minimum of our nitrogen

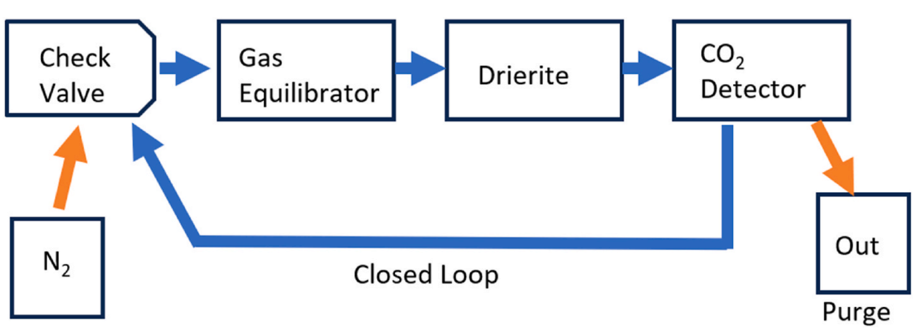

Fig. 1. Diagram of the coupling between a LICOR-850® (LI-850) CO_2 detector and a gas equilibrator (RAD-H_2O®, Durridge).

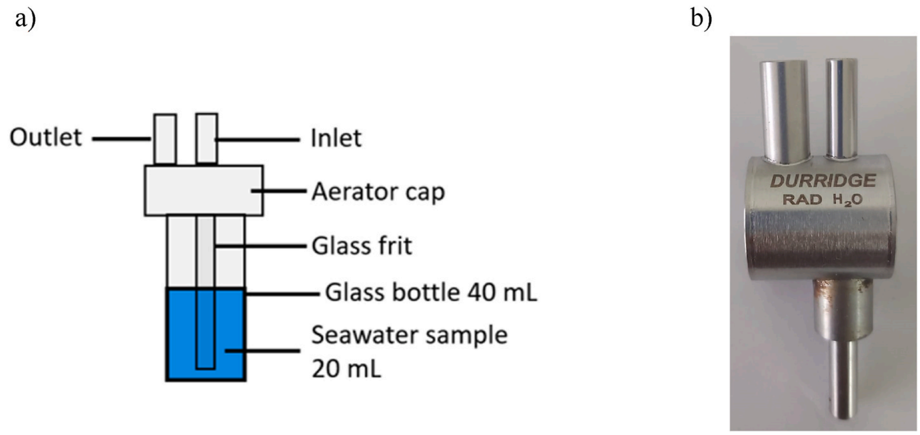

Fig. 2. (A) Scheme of the gas equilibrator (sample bottle). (b) Picture of the aerator cap.

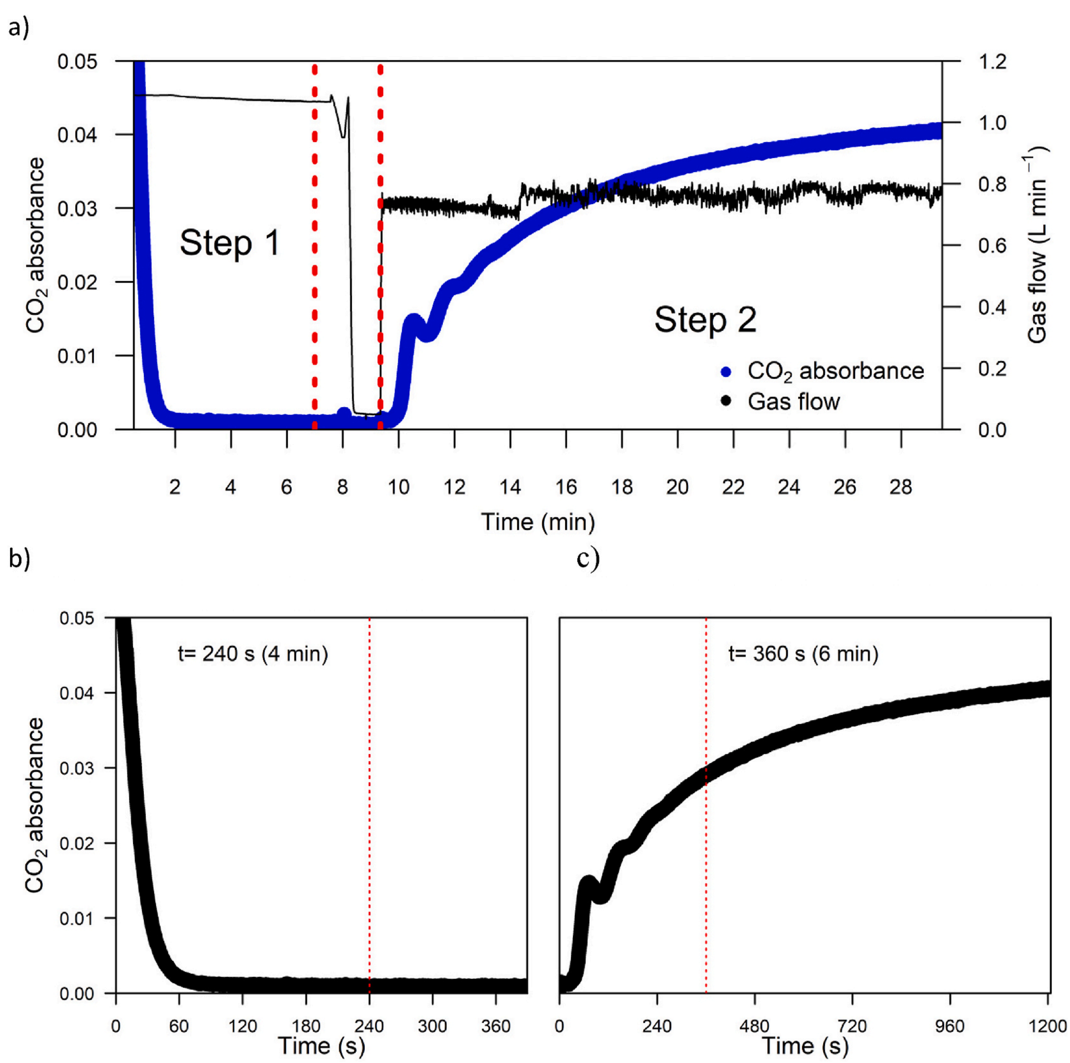

Fig. 3. Diagram of the analysis process of $p(CO_2)$ in a seawater sample: a) whole process, b) air evacuation step, c) sample measurement step.

regulator and maximum allowed by the LI-850). The flow was zero during the configuration change and ~0.7 L min^{-1} (LI-850 pump) during measurement.

The CO_2 record shows the two steps of the measurement process. During air evacuation (Fig. 3a, step 1), the air CO_2 concentration rapidly decreased during its displacement by $N_{2(g)}$. During measurement (Fig. 3a, step 2), CO_2 concentration increased following a saturation-type behavior due to the extraction of seawater-dissolved CO_2. The data between the red dotted line (Fig. 3a) corresponds to the configuration change from the open system to the closed loop (not considered for data analysis).

We calculated the system blank as the average CO_2 absorbance during the air evacuation. In our experimental setup, stability was observed between 4 and 6 min (Fig. 3b, data points after the red dotted line). During CO_2 homogenization, the concentration oscillated (Fig. 3c; before the red dotted line). Saturation was not reached because the carbonate system is continuously altered during measurement (DIC, pH, temperature, and salinity).

The data recorded during 6 and 20 min (CO_2 *Abs*) were fitted to a generalized saturation-type curve (Fig. 4). The fit was performed using a non-linear regression method using the minpack.lm package (Elzhov et al., 2023) in R (R Core Team, 2022) within the RStudio environment (RStudio Team, 2020). The fitting process yielded the saturation absorbance value (Abs_{sat}), its associated uncertainty, and the values and uncertainties of the parameters b and c (Equation (3)).

$$Abs = Abs_{sat}\left(1 - e^{-b*(time-c)}\right) \tag{3}$$

2.2. Reference measurement of $p(CO_2)$

DIC was measured using an Automated Infrared Inorganic Carbon Analyzer (AIRICA®, Marianda). Briefly, seawater was acidified with H_3PO_4 (8 %) to shift the carbonate system equilibrium CO_2 (Yin et al., 2020), purged with 99.9 % N_2, and directed towards a CO_2 detector (LI-7000) (Peterson et al., 2003) for analysis. The pH was measured by adding meta cresol purple indicator (P-mC) to seawater and measuring the absorbance spectrum using a UV–Vis spectroscopy system (Agilent 8453). The absorbances at different wavelengths (434, 578, and 730 nm) were used to calculate pH (Gómez et al., 2021).

DIC is the sum of the concentrations of the carbonate species (CO_2, HCO_3^- and CO_3^{2-}), so only a fraction (α_{CO_2}) corresponds to the CO_2 concentration ($[CO_2]$) (Equations (4) and (5)). This fraction can be determined using the concentration of hydrogen ions and the acidity constants of the carbonate system K_{a1} (corresponding to equilibrium in Equation (1)) and K_{a2} (corresponding to equilibrium in Equation (2)). These acidity constants were corrected with the salinity and temperature of each sample (Dickson et al., 2007):

$$\alpha_{CO_2} = \frac{1}{1 + \frac{K_{a1}}{[H^+]} + \frac{K_{a1}*K_{a2}}{[H^+]^2}} \tag{4}$$

$$[CO_2] = DIC^*\alpha_{CO_2} \tag{5}$$

The $p(CO_2)$ values and uncertainties were determined using the Seawater Carbonate Chemistry R package (*seacarb*; Gatusso et al., 2022), where $p(CO_2)$ is the ratio of CO_2 concentration to the Henry's constant (K_0, Equation (6)) corrected with each sample salinity and temperature (Dickson et al., 2007).

$$p\left(CO_2\right) = \frac{[CO_2]}{K_0} \tag{6}$$

2.3. Validation

To compare the two methodologies and assess the quality of the results, we analyzed blank solutions to perform a calibration curve, a substandard, and a certified reference material (CRM) used for oceanic CO_2 measurements (Dickson Batch 151; Dickson, 2015). Although this CRM is not certified for $p(CO_2)$, we used the certified total alkalinity and DIC values to estimate $p(CO_2)$ (Kim et al., 2023).

2.3.1. Solutions

2.3.1.1. Carbonate-free water. Carbonate-free water was used to prepare the solutions described in this section. It was prepared by boiling deionized water (Hycel, CAS: 7732-18-5) at ~100 °C for 30 min (Corning® Stirrer and hot plate PC-220), followed by bubbling with nitrogen gas (Linde, 99.9 %) for 10 min. The carbonate-free water was stored in 250 mL amber polypropylene containers at room temperature and sealed with Parafilm paper until use.

2.3.1.2. Carbonate-free synthetic seawater. The carbonate-free synthetic seawater was a solution of 0.72 M NaCl (JT Baker 99 %, CAS: 7647-14-5) in carbonate-free water. Its ionic strength is equivalent to the average seawater salinity (35 PSU; Millero, 2010). The solution was stored at room temperature in 250 mL borosilicate bottles with plastic caps, sealed with parafilm.

2.3.1.3. Calibration solutions. Four 250 mL standard solutions were prepared in borosilicate bottles with plastic caps, sealed with parafilm, and kept at room temperature. The concentrations of reagents used are listed in Table 1. To prepare each solution, sodium chloride (NaCl, Sigma-Aldrich, ~99 %, CAS: 7647-14-5) was first added to each bottle to

Table 1
Solution concentrations used for the $p(CO_2)$ calibration curve.

Solution	1	2	3	4
NaCl (mol L^{-1})	0.7271	0.7241	0.7211	0.7185
$NaHCO_3$ (mol L^{-1})	0.0029	0.0058	0.0087	0.0120
Total Alkalinity (μmol kg^{-1})	3584 ± 16	7143 ± 33	10,654 ± 50	14,531 ± 66
$p(CO_2)$ (μatm)	279 ± 5	479 ± 9	706 ± 14	1030 ± 20

Sodium bicarbonate ($NaHCO_3$, Sigma-Aldrich, ~99 %, CAS: 144-55-8) and sodium chloride (NaCl, Sigma-Aldrich, ~99 %, CAS: 7647-14-5).

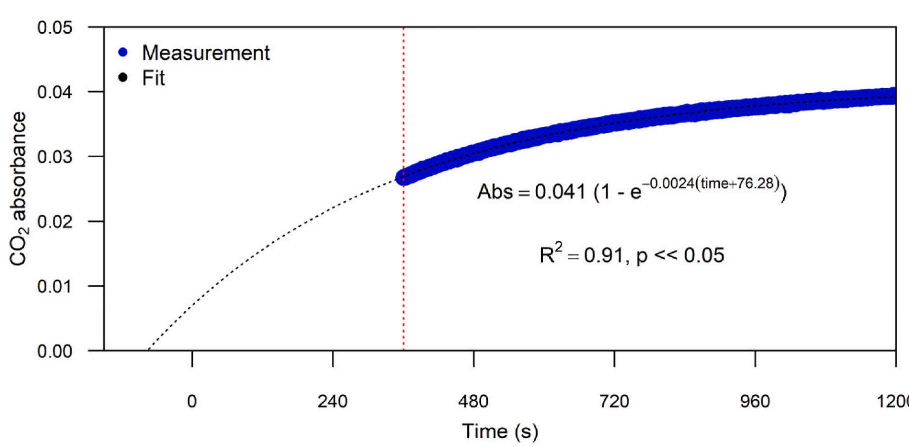

Fig. 4. Example of a non-linear regression fit to a measurement of $p(CO_2)$ in a real sample.

achieve a final ionic strength of 0.72 M, and 200 mL of carbonate-free water was added to dissolve the salts. Then, sodium bicarbonate ($NaHCO_3$, Sigma-Aldrich, ~99 %, CAS: 144-55-8) was added to adjust the $p(CO_2)$ levels. Finally, the total volume was adjusted by adding 50 mL of carbonate-free water. The reported $p(CO_2)$ values (Table 1) were measured with the reference method.

2.3.1.4. Substandard. A substandard was prepared according to the procedure outlined by Dickson to prepare a CRM without headspace control (NOAA, 2021), adapted to the available resources in the laboratory. Briefly, a submersible water pump was used to collect 5 L of seawater from Mazatlan Bay, NE Pacific. The recipient was constantly stirred for 24 h on an orbital shaker (Heidolph® Rotamax 120) to homogenize the sample and balance the CO_2 levels with those in the laboratory. The water was then filtered through a 0.7 µm glass microfiber filter (Whatman® CAT, 1825047) using a pump (Laboport ® N96). The filtered water was transferred with a peristaltic pump (Cole-Parmer ® Mini-pump with variable flow) to 1 L bottles with frosted lids. To each bottle, 0.8 mL of a $HgCl_{2(sat)}$ solution was added, and the bottles were sealed with carbon-free vacuum grease (Apiezon ® L Grease) and stored at room temperature.

2.3.2. Validation parameters

In this study, the direct CO_2 measurement with LI-850 was validated. To maintain a constant volume, all gas equilibrator bottles contained 20 mL of the reagent to be measured. To further assess the method's robustness across a broader salinity range, a calibration curve was developed using six solutions with a constant $NaHCO_3$ concentration (0.0116 mol L^{-1}) but varying salinities (0.582, 5.000, 15.000, 25.000, 35.000, and 45.000 PSU, uncertainty 0.003 PSU). The $p(CO_2)$ was measured using the direct method, and the results were analyzed to determine the salinity range where the original calibration (developed at 35 PSU) remains valid. A statistical analysis was performed to determine the salinity threshold below which a new calibration is needed. The reference average $p(CO_2)$ was calculated from the three highest salinity points (25, 35, and 45 PSU), and the $p(CO_2)$ values at lower salinities were compared to this reference using the measurement uncertainties.

2.3.2.1. Detection and quantification limits. To evaluate the Detection Limit (DL) and Quantification Limit (QL), $p(CO_2)$ in ten carbonate-free synthetic seawater samples was measured using the LI-850 direct method and the reference method. Equations (7) and (8) were used to determine the limits:

$$DL = \mu + 3 * \frac{\sigma}{\sqrt{n}} \tag{7}$$

$$QL = \mu + 10 * \frac{\sigma}{\sqrt{n}} \tag{8}$$

where μ is the average, σ is the standard deviation, and n is the number of data (10 assays per method).

2.3.2.2. Repeatability. To evaluate repeatability, we analyzed 16 bottles of the substandard using both methods (8 for each technique) on the same day. The standard deviation was calculated to determine its variability.

2.3.2.3. Accuracy and reproducibility. A total of 8 CRM samples were analyzed on different days over two months. Reproducibility was evaluated through the standard deviation. Accuracy was assessed using Equation (9) (Magnusson and Örnemark, 2014):

$$Accuracy\ (\%) = \frac{p(CO_2)_{average}}{p(CO_2)_{reported}} * 100 \tag{9}$$

where $p(CO_2)_{.average}$ is the average $p(CO_2)$ measured with the LI-850, and $p(CO_2)_{.reported}$ is the $p(CO_2)$ calculated from the certified DIC and TA, which was 632 ± 18 µatm.

2.3.2.4. Robustness. To estimate the method's robustness (Magnusson and Örnemark, 2014), we evaluated the effects of salinity and the laboratory room temperature on the analyses. Solution 2 from the calibration curve (Table 1, $p(CO_2)$ 479 ± 7 µatm) was used as standard prepared at two ionic strengths (0.68 and 0.72), equivalent to 33 and 35 PSU salinities, respectively. Both standards were analyzed (n = 3 each) at 20 °C and 25 °C by regulating the air conditioning system in the laboratory. A Pareto diagram was developed to determine the effects of these variables on the measurement of CO_2 absorbance.

2.4. Calibration

We performed a regression analysis between the saturation absorbance (Abs_{sat}) measured with LI-850 and the $p(CO_2)$ ($p(CO_2)_{calculated}$) derived from the reference method (Table 1), from which $p(CO_2)$ can be calculated for any saturation absorbance within the calibration range (0–1030 ppm) (equation (10)).

$$p(CO_2)_{calculated} = 51933(Abs_{sat})^2 + 6915(Abs_{sat}) - 3.41 \tag{10}$$

2.5. Study area

The $p(CO_2)$ methods were compared in coastal waters from the Coastal Observatory of Global Change in Mazatlan, Mexico (Sanchez-Cabeza et al., 2019). Seawater samples were collected along a transect from southern Mazatlan Bay (3 samples) and Estero de Urias Lagoon (5 samples) (NE Pacific; Fig. 5). Stations 4–8 are significantly impacted by anthropogenic activities, such as the Mazatlan port, urban wastewater and runoff, the fish industry, and a thermoelectric power plant, although mangroves progressively surround them (Cardoso-Mohedano et al., 2016a).

The samples were collected on 2022-05-09 following the procedure recommended by the Research Network of Marine-Coastal Stressors in Latin America and the Caribbean (REMARCO; Sánchez-Noguera, 2021). Samples were collected at a depth of 0.5 m using a horizontal Van-Dorn bottle (WILDCO, a 3.2 L horizontal acrylic bottle) and transferred to two separate bottles, one for each method. We used 125 mL borosilicate bottles for the reference method and 250 mL borosilicate bottles for the direct method. To prevent CO_2 exchange with the atmosphere, a Tygon tube was placed at the bottom of the bottle and filled with three times the bottle volume. For the reference method measurements, the borosilicate bottles were sealed with a frosted lid and vacuum grease (Apiezon® L Grease), whereas for the direct method, septum caps were used.

3. Results

3.1. Direct CO_2 measurement

The reproducibility and accuracy of the direct method were 12 % and 91 %, respectively (Fig. 6a), while repeatability (Fig. 6b) was 9 %. The QL (Fig. 6c) was 20.7 µatm, which is ~20 times lower than current atmospheric concentrations. The calibration curve (Fig. 6d) exhibited linear behavior with a high correlation coefficient (r = 0.998, p < 0.05), based on three measurements for each solution. The Pareto diagram (Fig. 6e) showed a significant effect of temperature on the $p(CO_2)$ analysis but not on salinity.

To further investigate the effect of salinity, a calibration was performed using solutions with a constant $NaHCO_3$ concentration (0.0116 mol L^{-1}) across salinities of 0.582, 5.000, 15.000, 25.000, 35.000, and 45.000 PSU (uncertainty 0.003 PSU). The $p(CO_2)$ values ranged from

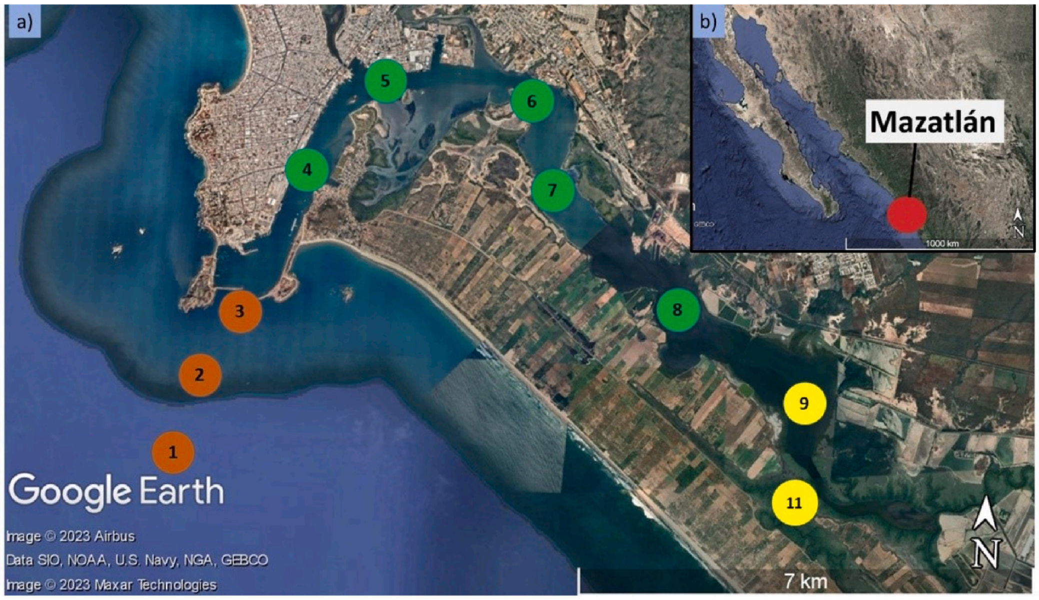

Fig. 5. Location of sampling stations from southern Mazatlan Bay (1–3) and Estero de Urias Lagoon. The inset shows the location of Mazatlán (23°14′57.89 ″N, 106°24′40.11″ W) at the entrance of the Gulf of California.

848 ± 59 µatm to 1183 ± 83 µatm PSU (Fig. 7). A reference average $p(CO_2)$ from the three highest salinity points (25, 35, and 45 PSU) was 1174.7 µatm with a standard error of 27.3 µatm. To test the statistical significance of the difference, we calculated its standard error by quadratic propagation. The $p(CO_2)$ value at 15 PSU (1119 ± 78 µatm) was not significantly different from the reference, while the values at 5 PSU (1000 ± 70 µatm) and 0.582 PSU (848 ± 59 µatm) showed significant deviations (differences = 174.7 µatm and 326.7 µatm, respectively). Therefore, the method is reliable for salinities down to 15 PSU, but a new calibration is needed below this threshold.

3.2. Analytical comparison between methodologies

To compare both methodologies, we conducted the same experiments to verify the reference method (DIC and pH), and the results were used to calculate $p(CO_2)$ (Table 2). The results showed that the direct measurement had a higher blank $p(CO_2)$ value than the indirect measurement with the reference method. This could be attributed to some air input when the switch transitions from an open to a closed condition, or to minor air leakage during measurement, as laboratory air contains CO_2. As a result, both the DL and QL were higher for the direct measurement than for the reference method. Furthermore, the repeatability and reproducibility of the direct measurement method were also larger than those of the reference method. This is reflected in the uncertainty of the CRM direct measurement, which was larger (37 µatm) than that of the indirect measurement (18 µatm), although both methods demonstrated good accuracy.

3.3. Analysis of real samples

The $p(CO_2)$ values of stations 1–8 (Table 3, Fig. 8) were comparable (Student's t-test, $p > 0.05$) between the two methodologies (range 360–700 µatm, average 528 ± 48 µatm for the direct method, and 400–886 µatm, average 613 ± 31 µatm for the reference method). For both methodologies, the $p(CO_2)$ values are lower at the offshore stations 1–3 (407 ± 37 µatm and 435 ± 22 µatm for direct and reference measurements, respectively).

4. Discussion

4.1. Analytical comparison

First and foremost, the proposed direct method shows excellent linearity ($r^2 = 0.998$). The salinity calibration confirmed that the method is robust for salinities as low as 15 PSU, where $p(CO_2)$ values remain consistent with the high-salinity reference within measurement uncertainty. Below 15 PSU, significant deviations were observed, indicating the need for a new calibration in lower salinity environments, such as brackish waters. This is likely due to changes in CO_2 solubility and gas-liquid partitioning at lower ionic strengths, which affect headspace equilibration in the closed-loop system. While this methodology is robust for typical analytical temperatures (20–25 °C) and seawater salinities (15–45 PSU), its application in brackish waters requires further validation and calibration adjustments (see Section 4.4).

The observed reproducibility (10 %) and repeatability (7 %) of the direct method, while acceptable for environmental applications in dynamic coastal waters, are notably lower than those of the reference method. Several practical improvements could enhance the precision of the direct measurement system. Implementing an automated control of equilibration time and sample switching (e.g., with electronic valves) could improve consistency across runs and reduce operator-induced variability. The integration of temperature-controlled enclosures for both the sample and the detector could minimize fluctuations in CO_2 solubility and gas-phase concentration during measurements. Refining the nitrogen flushing protocol and using higher-purity gas or preconditioning steps could help stabilize the system blank and reduce its contribution to overall uncertainty. Finally, developing a calibration routine using internal standards or synthetic reference waters may improve inter-day consistency and facilitate long-term quality control. These enhancements would not only improve measurement precision but also increase the robustness of the method for multi-user operation.

The limit of quantification of both methods is much lower than expected $p(CO_2)$ values in surface seawater, and the direct measurement method also shows good accuracy (91 %), indicating that it is reliable for measuring $p(CO_2)$ in seawater samples. Our results indicate that the direct method is reliable and suitable for analyzing $p(CO_2)$ in water with a salinity of ~35, pH ~8.1, and alkalinity of ~2366 µmol kg^{-1}. However, each laboratory should use this information cautiously, depending

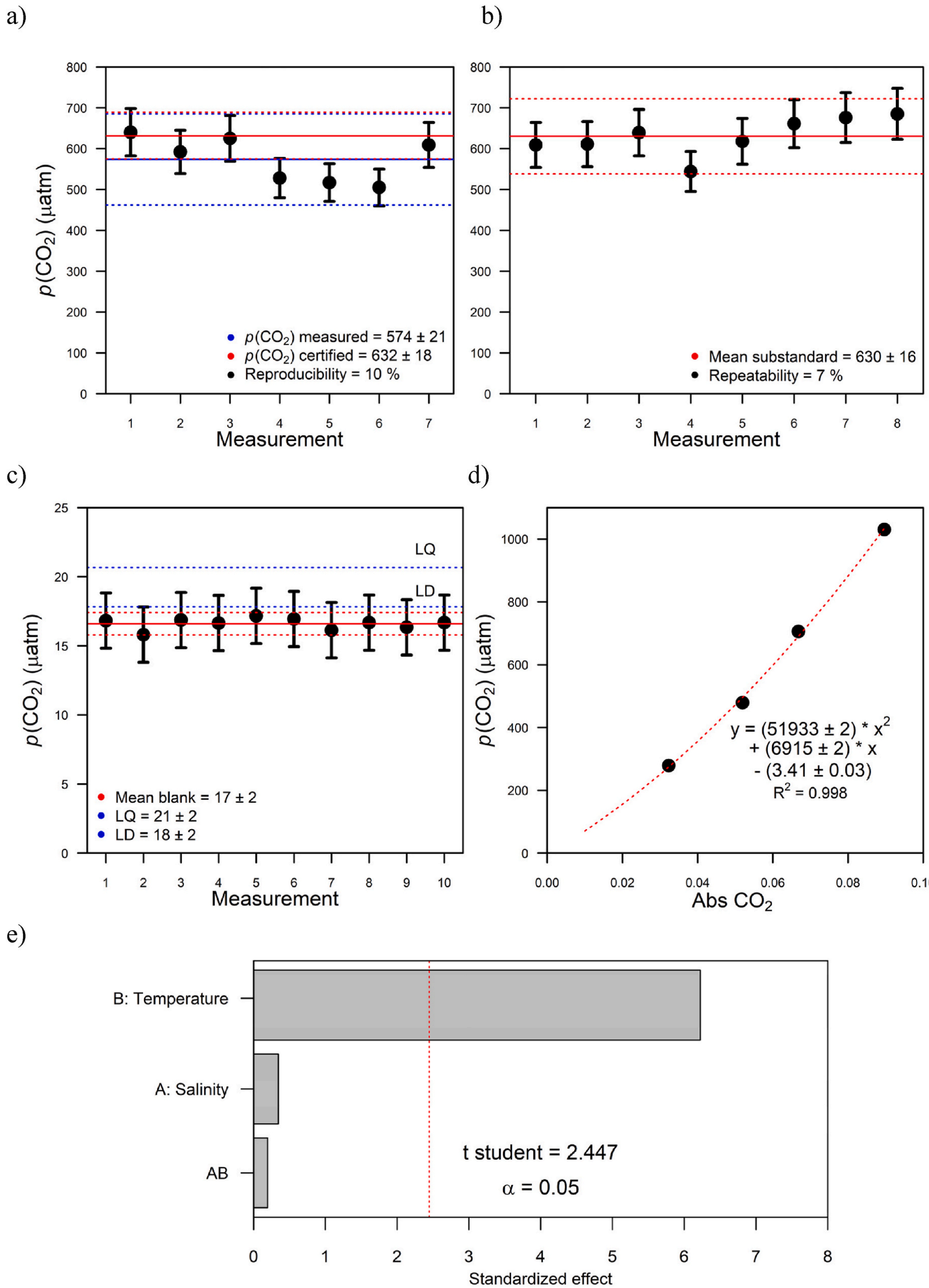

Fig. 6. Verification of the direct measurement methodology. a) Reproducibility and accuracy (Shewhart diagram), b) substandard repeatability, c) average blank, d) calibration curve, and e) Pareto chart.

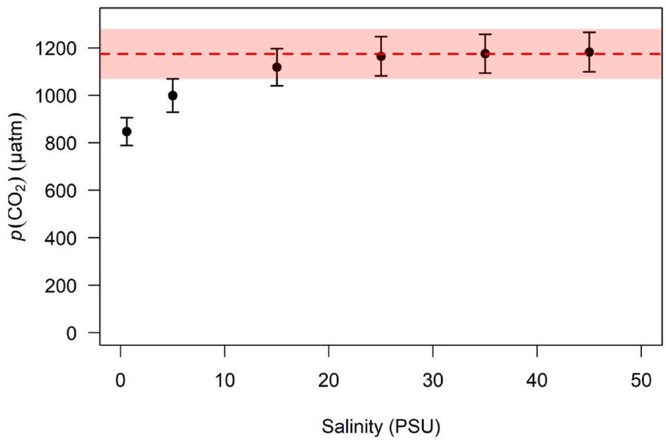

Fig. 7. Salinity vs. $p(CO_2)$ calibration curve with a constant $NaHCO_3$ concentration (0.0116 mol L^{-1}). The reference average $p(CO_2)$ (1174.7 µatm) is shown as a dashed red line, with the threshold (2 x standard error of the difference to the reference value) as a shaded area.

Table 2
Summary of the validation exercises.

$p(CO_2)$ (µatm)	Indirect measurement (DIC + pH)	Direct measurement
Blank	4.22 ± 0.07	16.6 ± 0.1
Detection Limit (DL)	5.0 ± 0.1	18 ± 2
Quantification Limit (QL)	6.9 ± 0.1	21 ± 2
Repeatability	0.4 %	7 %
Reproducibility	2.3 %	10 %
CRM ($p(CO_2)$ measured)	632 ± 18	574 ± 21
Accuracy	99 %	91 %

Table 3
Comparison of $p(CO_2)$ results in eight surface seawater samples from southern Mazatlan Bay (stations 1–3) and Estero de Urias Lagoon (stations 4–8) (Fig. 5) on 2022-05-09. Other carbonate system variables are provided.

Station	$p(CO_2)$ Direct (µatm)	$p(CO_2)$ Indirect (µatm)	DIC (µmol kg^{-1})	pH	Total alkalinity (µmol kg^{-1})
Mazatlan Bay					
1	360 ± 32	452 ± 8	2106 ± 9	8.019 ± 0.003	2383 ± 7
2	411 ± 37	407 ± 7	2093 ± 9	8.073 ± 0.003	2362 ± 7
3	450 ± 40	460 ± 8	2100 ± 9	8.016 ± 0.003	2359 ± 6
Estero de Urias Lagoon					
4	495 ± 45	772 ± 14	2200 ± 9	7.827 ± 0.003	2363 ± 5
5	611 ± 55	892 ± 16	2218 ± 9	7.768 ± 0.003	2360 ± 4
6	629 ± 57	674 ± 12	2202 ± 9	7.879 ± 0.003	2400 ± 5
7	570 ± 51	518 ± 9	2108 ± 9	7.966 ± 0.003	2353 ± 6
8	700 ± 63	775 ± 14	2177 ± 9	7.816 ± 0.003	2344 ± 5

on the characteristics of the ecosystems of interest.

Although the direct method exhibited higher blank values (16.6 µatm) and detection/quantification limits (18 and 21 µatm, respectively) than the reference method, these differences are not critical in practical applications for typical coastal waters. Surface seawater $p(CO_2)$ values generally range from 200 to over 1000 µatm in dynamic coastal systems, as observed in this study. Therefore, both methods provide sufficient sensitivity for detecting relevant variations in $p(CO_2)$ under natural conditions, especially in coastal and estuarine environments, where $p(CO_2)$ is typically elevated due to biological and anthropogenic influences. However, the elevated blank in the direct method may introduce a minor positive bias when analyzing very low $p(CO_2)$ samples, such as those from oligotrophic open-ocean waters or artificially equilibrated solutions. Further refinement to reduce blank contributions (e.g., through improved sealing or pre-conditioning protocols) could enhance its applicability in very low $p(CO_2)$ environments.

4.2. Sample analysis

The study area shows a wide range of environments. Stations 1–3 correspond to coastal waters, and anthropogenic activities have a significant influence on stations 4–8. The average $p(CO_2)$ value at the coastal stations (1–3) (407 ± 37 µatm) is similar to the global atmospheric concentration (421 µatm as of May 2022; Stein, 2022), indicating that atmospheric exchange is the primary source of CO_2 in this area. However, the $p(CO_2)$ values observed in the lagoon stations 4–8 (average of 618 ± 56 atm) are higher than in coastal stations, indicating the influence of anthropogenic activities of Mazatlán city (Montaño-Ley et al., 2008). Since the decomposition of organic matter produces CO_2, we hypothesize that $p(CO_2)$ levels are influenced by the discharges of organic matter from the food industry, wastewater treatment plants, and the impact of urban rainwater and wastewater through the Estero del Infiernillo (Cardoso-Mohedano et al., 2015, 2016b; Rios-Mendoza et al., 2021).

Although the direct and indirect $p(CO_2)$ measurements were statistically comparable overall, significant discrepancies at Stations 4 and 5 (direct method: 277–281 µatm lower) align spatially with intense anthropogenic pressures near the Estero del Infiernillo outfall. These sites exhibited high biological signatures (Table 4: Chlorophyll-*a* 9.2–16.7 µg L^{-1}) 16–28 × higher than offshore stations, consistent with nutrient enrichment from wastewater/fish processing discharges. This organic matter load likely promoted microbial respiration, elevating dissolved CO_2 while simultaneously introducing surfactants and turbidity that may have impeded gas exchange efficiency during direct method equilibration or affected the reference method. The direct approach's sensitivity to transient CO_2 pulses from rapid decay could further explain its lower values versus the integrative indirect method. While controlled experiments are needed to isolate mechanisms, the co-occurrence of maximal $p(CO_2)$ differences with peak anthropogenic indicators underscores the importance of validating novel methods in dynamic, impacted systems where biogeochemical disequilibria are prevalent.

Furthermore, although both sampling methods aimed to minimize atmospheric exchange, small differences in sampling timing, handling, or headspace equilibrium may have affected the direct method, which is likely more sensitive to transient disequilibria in high-$p(CO_2)$ waters. Additionally, bubbles, high turbidity, or surfactants at these sites could have altered gas exchange efficiency during headspace equilibration, potentially leading to underestimation in the direct method. Further controlled studies under varying estuarine conditions are required to better understand these discrepancies. However, the results highlight the importance of validating direct $p(CO_2)$ methods in diverse and dynamic coastal environments, particularly where strong biogeochemical gradients are expected.

4.3. Operational comparison

The measurement of $p(CO_2)$ in seawater has numerous and important applications, especially for quantifying the role of oceans and coastal areas as sinks or sources of CO_2 at a global level (IPCC, 2021). However, the operating cost can make it unfeasible for many laboratories. This study presents a new methodology for analyzing $p(CO_2)$ reliably and affordably for many laboratories.

While the underlying principles of gas equilibration and NDIR detection are established, the novelty of our method lies in their synergistic integration and optimization into a simple and cost-effective system for seawater $p(CO_2)$ measurements. Traditional equilibration methods often rely on open systems or large-volume equilibrators, which require substantial gas volumes and are prone to contamination from ambient air, particularly problematic in dynamic field

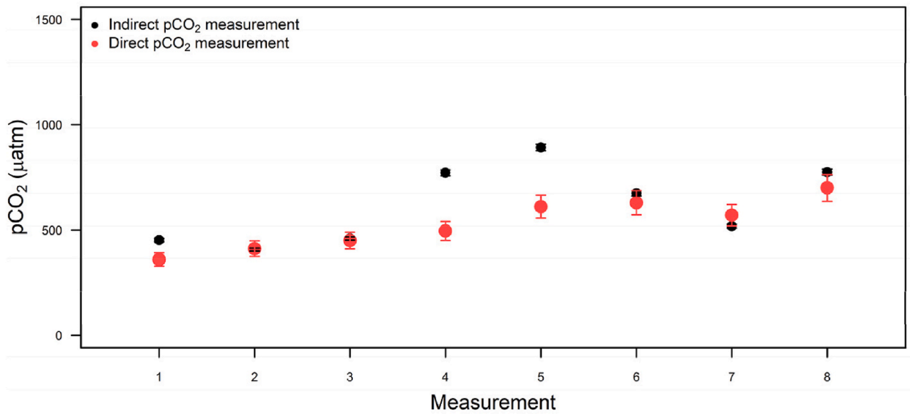

Fig. 8. Comparison of $p(CO_2)$ values, measured with the direct and reference (indirect) methodologies, in surface seawater samples from stations in the southern Mazatlan Bay (stations 1–3) and Estero de Urias Lagoon (stations 4–8), collected on 2022-05-09.

Table 4
Environmental variables of surface (0.5 m) water samples from Mazatlán Bay (stations 1–3) and Estero de Urías Lagoon (stations 4–8) on 2022-05-09.

Station	T (°C)	±	Salinity	±	DO saturation (%)	±	Chl *a* (µg L^{-1})	±	$p(CO_2)$ (µatm)	±
1	25.32	0.01	33.0	0.1	109.6	0.1	0.99	0.10	360	32
2	25.42	0.02	28.2	0.1	114.8	3.1	0.97	0.60	411	37
3	25.59	0.03	31.0	0.1	111.8	2.2	0.82	0.11	450	40
4	25.84	0.02	30.3	0.1	105.5	1.1	9.2	7.1	495	45
5	28.21	0.07	31.0	0.1	115.0	3.8	16.7	8.3	611	55
6	30.12	0.05	31.9	0.1	156.7	10.6	28.8	12.3	629	57
7	29.89	0.07	32.9	0.1	158.0	10.2	13.5	5.4	570	51
8	29.50	0.15	32.1	0.1	101.2	0.7	2.6	0.6	700	63

environments. Our closed-loop system, utilizing a small 40 mL gas equilibrator, recirculates a minimal volume of nitrogen gas through the seawater sample. This design ensures that the headspace CO_2 concentration equilibrates with the dissolved CO_2 without the need for a continuous gas supply or large equipment. By minimizing gas consumption and reducing contamination risks, this approach is especially advantageous in coastal field settings, where maintaining a controlled environment is challenging. The closed-loop configuration also eliminates logistical complexities, such as transporting large gas cylinders, making it highly practical for remote or resource-limited locations.

The incorporation of a NDIR detector into the closed-loop system allows for direct measurement of CO_2 in the equilibrated headspace. This avoids the need for indirect calculations based on two carbonate system parameters (e.g., pH and DIC). Traditional indirect methods introduce complexity and potential errors due to parameter interdependencies and equilibrium assumptions, particularly in dynamic coastal waters. Our direct approach simplifies the measurement process, reduces data processing time, and enhances reliability, making it ideal for real-time field monitoring.

The integration of these components significantly reduces costs compared to conventional methods (Table 5). Traditional $p(CO_2)$ measurement often requires expensive instruments like UV–Vis spectrophotometers (for pH) and automated carbon analyzers (for DIC), with combined costs exceeding USD 90,000. In contrast, our system utilizes a commercially available NDIR detector (LI-850, approximately $7600 USD in 2022) and a small, simple gas equilibrator, which significantly reduces initial equipment costs by over 90 %. Additionally, the closed-loop design minimizes operational expenses by reducing the need for reagents (e.g., acids, indicators) and carrier gases. For example, only a small volume of nitrogen is required during the initial air evacuation phase, compared to the continuous gas flow demanded by many traditional setups. This cost-effectiveness could allow laboratories in developing regions or with limited budgets to participate in critical ocean acidification and carbon cycle research by direct monitoring of $p(CO_2)$ in

Table 5
Operational comparison of the direct (LICOR-850) and reference (UV–VIS and AIRICA) methods to determine $p(CO_2)$.

	Reference method	Direct measurement
Analysis time	96 min x sample 2 analysis	28 min x sample 1 analysis
Reagents	Nitrogen Drierite Phosphoric acid meta-Cresol Purple (mCP) TRIS Dickson's water	Nitrogen Drierite
Equipment	AIRICA LICOR-7000 UV–Vis spectrophotometer	LICOR-850 Gas equilibrator
Approx. cost of equipment (2022)	93,500 USD	7600 USD
Portability	No	Yes

seawater samples. From an operational standpoint, it is important to consider factors such as human resources (analysis time), infrastructure (equipment), reagents, and portability of the methodology to be used in field conditions (Table 5). In comparison to the reference method, the direct method is faster (28 min per sample), requires a smaller quantity of reagents, and incurs 12 times lower equipment costs ($ 7600 against USD 93,500 in 2022) (Table 5). The most important cost of the methodology is the LICOR-850 CO_2 detection system. We believe that the proposed methodology could be reduced in cost by utilizing a low-cost NDIR detector and cost-effective materials.

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For the reference method, it is typical to have two or more analysts to obtain results quickly, whereas only one analyst is needed for the direct method. In addition, the direct method can be implemented for in situ measurements (in the sampling sites or even an oceanographic vessel) since the LI-850 is compact, robust, and lightweight. In contrast, the AIRICA and the UV–Vis spectrometer are generally delicate and heavy and require tight environmental control during measurements.

The proposed method fills a gap in $p(CO_2)$ monitoring by offering a practical solution for laboratories with limited financial and technical resources. Traditional methods, while accurate and widely used, are expensive and often confined to well-funded research institutions with access to specialized instruments, trained personnel, and climate-controlled laboratories. This creates a disparity in the spatial and temporal resolution of $p(CO_2)$ data, particularly in coastal regions from developing countries. By substantially reducing costs and simplifying operations, the proposed method enables broader implementation of $p(CO_2)$ measurements, including in field stations or mobile observatories. Its portability and operational simplicity make it feasible to integrate into long-term monitoring programs, citizen science efforts, and collaborative regional networks (e.g., REMARCO, www.remarco.org), thereby contributing to a more extensive observation of ocean acidification and coastal carbon dynamics.

4.4. Potential adaptation to variable estuarine conditions

Although this study focused on typical coastal seawater conditions (salinity ~35 PSU, pH ~8.1), the method may be used in other environments such as estuaries, coastal lagoons, and riverine plumes, where salinity and pH can fluctuate rapidly over short temporal and spatial scales. These environments often challenge the applicability of indirect $p(CO_2)$ estimation methods due to uncertainties in carbonate system equilibrium constants under low salinity and variable ionic strength. In contrast, the direct measurement approach proposed here does not rely on such equilibrium assumptions and is thus, in principle, more flexible.

However, further validation is required to ensure accurate equilibration and gas-phase stability in waters with low alkalinity, high turbidity, or low buffer capacity. For instance, the kinetics of CO_2 exchange may differ in brackish water, potentially affecting the time required to reach saturation in the closed-loop system. Differences in sample matrix (e.g., organic matter, particulate load) may influence gas-liquid partitioning or interfere with NDIR sensor readings. Future work should explore these scenarios using controlled synthetic brackish solutions and in situ estuarine sampling to assess performance and refine the method accordingly. Such adaptations could enable broader deployment of direct $p(CO_2)$ monitoring across complex coastal gradients, which are currently underrepresented in global carbon budgets.

Salinity influences the equilibrium constants (e.g., K_{a1} and K_{a2} for carbonic acid dissociation) and CO_2 solubility (K_0) (Dickson et al., 2007). In brackish waters, with reduced ionic strength, these parameters shift, affecting the relationship between $p(CO_2)$ and the measured saturation absorbance. To enable accurate $p(CO_2)$ measurements in brackish waters (<15 PSU), we recommend the following recalibration protocol.

– Create synthetic seawater solutions with salinities below 15 PSU using sodium chloride and sodium bicarbonate ($NaHCO_3$) to simulate a suitable range of $p(CO_2)$ levels, depending on expected salinities. Ideally, these solutions should mimic the ionic composition of the brackish waters.
– Establish reference $p(CO_2)$ values by measuring the "true" $p(CO_2)$ of each solution using a reference method, such as indirect calculation from pH and dissolved inorganic carbon (DIC) or total alkalinity, under controlled laboratory conditions.
– Measure with the proposed method to determine the saturation absorbance for each solution.
– Generate a calibration curve (measured absorbance vs. reference $p(CO_2)$) to create a salinity-specific calibration curve.
– Validate with real brackish water samples, comparing results with the reference method to confirm accuracy.

Temperature also affects CO_2 solubility (K_0) and the carbonate system's equilibrium constants (K_{a1}, K_{a2}), altering the partitioning of CO_2 between seawater and the headspace in the gas equilibrator (Dickson et al., 2007). Additionally, the NDIR detector's infrared absorbance measurements may vary with temperature due to changes in sensor sensitivity or gas-phase properties. In this work, the measurement temperature refers to the conditions under which the analysis is conducted in the laboratory (not the sample's original temperature). A temperature range of 20–25 °C is typically used, as it reflects standard laboratory conditions that ensure proper maintenance and optimal performance of analytical instruments. Under other conditions (such as research vessels or coastal observatories), we outline a calibration procedure adapted to the expected temperature ranges to ensure accurate $p(CO_2)$ measurements.

– Prepare calibration solutions with varying $NaHCO_3$ concentrations to achieve $p(CO_2)$ levels covering the expected range (e.g., 200–1000 μatm, Table 1). Store the solutions in sealed borosilicate bottles until use, ensuring minimal CO_2 exchange with the atmosphere.
– Establish reference $p(CO_2)$ values by measuring the "true" value of each solution using a reference method, such as indirect calculation from pH and dissolved inorganic carbon (DIC) or total alkalinity, under controlled laboratory conditions.
– Define the expected temperature range for the laboratory or field setting, and select calibration temperatures that span this range to ensure robust coverage.
– Temperature control: In the laboratory, place the calibration solutions in a water bath or climate-controlled chamber set to the target temperature. Allow solutions to equilibrate for at least 30 min to ensure uniform temperature. Measure the saturation absorbance at each temperature for each solution using the proposed method.
– Generate calibration curves (measured absorbance vs. reference $p(CO_2)$) to create temperature-specific calibration curves.
– Validate with real brackish water samples, comparing results with the reference method to confirm accuracy.

4.5. Limitations and future directions

Despite the advantages of the proposed method (such as lower cost, portability, and good analytical performance), several limitations must be acknowledged. It relies on a specific commercial CO_2 detector (LI-850), which, while more affordable than traditional systems, still is a significant initial cost for some laboratories and introduces dependence on proprietary hardware and software. This dependency could be mitigated by adapting the method to alternative low-cost non-dispersive infrared (NDIR) sensors, which are increasingly available for environmental monitoring and may further reduce implementation costs while maintaining adequate analytical performance.

Cost-effective technologies are vital for scaling up environmental monitoring, particularly in coastal regions where understanding carbon dynamics requires extensive spatial and temporal data. Low-cost NDIR sensors, if adapted effectively, could ease access to $p(CO_2)$ measurements, enabling broader research participation and enhancing scalability. The field of gas sensing offers several commercially available options designed for CO_2 detection, primarily in air. These sensors can potentially be adapted for seawater $p(CO_2)$ measurements with modifications, such as integration into a closed-loop equilibration system like the one described here. For example, SenseAir Sunrise AB (<100 USD), this compact, low-power NDIR sensor has a measurement range of 0–5000 ppm and an accuracy of approximately ±30 ppm (±3 % of reading). Its affordability and precision make it a strong candidate for adaptation to seawater applications.

Adapting these sensors for seawater $p(CO_2)$ measurements involves customizing the gas exchange interface and calibrating for seawater-specific conditions (e.g., temperature and salinity). While this adaptation may require additional effort, it is typically less costly than developing new sensors (e.g., ~100 USD for enclosures and calibration). Low-cost sensors are not inherently designed for the high humidity and variable conditions of seawater measurements, but with advanced calibration techniques (e.g., machine learning), low-cost NDIR sensors can achieve accuracies comparable to higher-end models in atmospheric applications (Dubey et al., 2024). Low-cost sensors often require less specialized training and maintenance, further lowering the total cost of ownership and enhancing accessibility for diverse research teams. Overall, the potential cost of a low-cost $p(CO_2)$ system might be more than one order of magnitude lower than our LI-850 system, therefore allowing access to many laboratories worldwide and serving local and regional environmental networks.

However, one must not forget that i) frequent recalibration or advanced correction methods may be necessary to maintain accuracy, adding complexity to large-scale use, ii) high humidity, temperature fluctuations, and biofouling in seawater environments could degrade performance, requiring robust enclosures and maintenance protocols, and iii) ensuring uniform quality across a sensor network demands rigorous quality control, such as periodic validation against reference instruments.

Since the method is based on headspace equilibration, it may be sensitive to operator handling, equilibration time, and gas-tightness, requiring careful standardization for inter-laboratory comparisons. Additionally, saturation equilibrium during the measurement phase is not fully achieved due to the dynamic alteration of the carbonate system during gas exchange, which may affect precision. The method exhibits a higher blank value and quantification limit compared to the reference method, likely due to residual atmospheric CO_2 in the system or minor leakage during the transition from an open to a closed configuration. This contributes to slightly higher uncertainties in measurements, particularly in samples with low concentrations.

Although the system showed robustness to typical variations in seawater salinity and laboratory temperature (20–25 °C), its performance under broader environmental conditions (such as brackish water or extreme temperatures) has not yet been evaluated. In this study, we validated the direct $p(CO_2)$ measurement method within a range of approximately 200 to 1000 μatm, which encompasses the typical conditions observed in our study area. We acknowledge that nearshore water bodies can exhibit higher $p(CO_2)$ levels, particularly in eutrophic or hypoxic environments. Although assessing the method's performance across this broader range is needed, this was out of scope of this first study, and future research is expected to focus on evaluating the method's reliability and accuracy across the full spectrum of $p(CO_2)$ levels encountered in nearshore waters. These limitations must be taken into account when applying the method, particularly for time-series observations or for environments with rapidly changing conditions.

The method's operational affordability makes it well suited for integration into regional and global long-term observation networks, such as REMARCO or the Global Ocean Acidification Observing Network (GOA-ON), especially in countries or coastal regions where traditional carbonate system measurements are limited by resource constraints. Coupling this method with analysis for dissolved oxygen, turbidity, or chlorophyll could also support multi-parameter characterization of biogeochemical processes in estuarine and coastal waters. These potential applications reinforce the method's relevance not only for improving local monitoring capacity but also for filling spatial gaps in global carbon flux assessments.

5. Conclusions

The measurement of $p(CO_2)$ in coastal waters is essential for assessing the role of coastal ecosystems as sources or sinks of atmospheric CO_2, the primary driver of recent climate change. The most commonly used technique for measuring $p(CO_2)$ involves mathematical calculation from two other carbonate system variables. In this study, we developed and validated a simple method to directly quantify $p(CO_2)$.

The proposed methodology consists of CO_2 extraction with nitrogen gas in a small gas equilibrator and direct measurement with a LI-850 CO_2 detector in a closed-loop configuration. During the air evacuation phase (6 min), the system baseline is determined, and during measurement (14 min), the saturation absorbance is calculated with a non-linear regression method. The methodology exhibits good accuracy (91 %), a low limit of quantification (~20 times lower than typical surface $p(CO_2)$ values), and excellent linearity, demonstrating its reliability for practical applications. It also proved robust under typical seawater conditions, confirming its potential for widespread use.

The novelty of this design lies in the innovative coupling of a closed-loop equilibration system with an NDIR detector, optimized for direct, accurate, and practical measurements of $p(CO_2)$ in coastal waters. This integration reduces costs by utilizing affordable technology and minimizing resource consumption, while simplifying field operations through portability, robustness, and ease of use. The system's reduced analysis time and operational simplicity make it a viable tool for high-frequency monitoring in dynamic coastal zones, where rapid environmental changes demand real-time data. Its affordability and ease of use also position it as an inclusive solution for resource-limited settings, enabling contributions to global datasets on coastal carbon dynamics. By lowering both financial and operational barriers, our method supports expanded participation in networks like REMARCO or GOA-ON, addressing critical gaps in $p(CO_2)$ data from underrepresented regions.

The field of low-cost NDIR sensors is evolving rapidly, offering options that, with adaptation, could measure seawater $p(CO_2)$. Their affordability and potential for calibration improvements make them promising for scalable monitoring. By reducing financial barriers, low-cost NDIR detectors could enable extensive coastal $p(CO_2)$ networks, advancing our understanding of carbon dynamics in a cost-effective manner.

The method is not intended to replace much-needed high-precision laboratory techniques, but rather to complement them by expanding the accessibility and frequency of measurements in dynamic and vulnerable environments. This method could become a valuable tool for researchers and environmental managers studying the carbon cycle and ocean acidification in coastal ecosystems, supporting informed decision-making and the development of mitigation strategies in the context of global change.

CRediT authorship contribution statement

Martín Rangel-García: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Joan Albert Sanchez-Cabeza:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Ricardo Adrián Martínez-Galarza:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Marcela Guillermina Fregoso-López:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Arturo García-Mendoza:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Ana Carolina Ruiz-Fernández:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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